ECN 594: Demographic Interactions and pyblp

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Plan for today

1. **Recap: The IIA problem**
2. The demographic interaction model
3. Why demographics help with IIA
4. Introduction to pyblp
5. Worked example: Basic logit
6. Worked example: With demographics
7. Interpreting results

Recap: The IIA problem

- Last time: basic logit gives elasticities

$$\eta_{jj} = \alpha p_{jt}(1 - s_{jt}), \qquad \eta_{jk} = -\alpha p_{kt} s_{kt}$$

- **Problem:** Cross-price elasticity η_{jk} doesn't depend on j!
- All products have the *same* cross-elasticity with product k
- Example: BMW raises price $\Rightarrow$ same fraction go to Mercedes as to Honda Civic
- This is unrealistic: BMW and Mercedes are closer substitutes
- **Today:** Fix this with demographic interactions

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The demographic interaction model

- We estimate a version of **Berry, Levinsohn & Pakes (1995)** — "BLP"

- Allow preferences to vary across consumers:

$$u_{ijt} = x_{jt}\beta_i + \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

- β_i, α_i: consumer i's taste for characteristics and price

- **Key question:** What drives heterogeneity in β_i and α_i?

- **Our approach:** Observed demographics

$$\beta_{ik} = \beta_k + \sum_{\ell=1}^{L} \gamma_{k\ell} D_{i\ell}, \qquad \alpha_i = \alpha + \sum_{\ell=1}^{L} \gamma_{\alpha\ell} D_{i\ell}$$

- $D_{i\ell}$: demographic ℓ for consumer i (income, age, family size, etc.)

Rewriting in terms of mean utility

- Substituting the expressions for β_{ik} and α_i:

$$u_{ijt} = \underbrace{x_{jt}\beta + \alpha p_{jt} + \xi_{jt}}_{\delta_{jt}\text{ (mean utility)}} + \underbrace{\sum_{k,\ell} \gamma_{k\ell} D_{i\ell} x_{jt,k}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- **Mean utility** δ_{jt}: part common to all consumers
  - Contains "**linear parameters**" (β, α) — enter linearly in δ
- **Individual deviation** μ_{ijt}: how consumer i differs from average
  - Contains "**nonlinear parameters**" $(\gamma_{k\ell})$ — enter nonlinearly via shares
- Note: $\alpha < 0$ (higher price $\rightarrow$ lower utility)

From utility to demand

- **Individual choice probability** (with T1EV errors):

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt} + \mu_{ikt})}$$

- **Market share** (aggregate over consumers):

$$s_{jt} = \frac{1}{I} \sum_{i=1}^{I} s_{ijt}$$

- This is **demand**: the fraction of consumers who buy product j

- Key: shares depend on mean utilities δ and demographics D_i

Examples of demographic interactions

- Recall: $\mu_{ijt} = \sum_{k,\ell} \gamma_{k\ell} D_{i\ell} x_{jt,k}$ captures preference heterogeneity
- Effective price coefficient for consumer i: $\alpha_i = \alpha + \gamma_{\text{inc}\times p} \cdot \text{income}_i$
- **Question:** High-income consumers are typically *less* price-sensitive.
  - Given $\alpha < 0$, what sign should $\gamma_{\text{inc}\times p}$ have?

Where do demographics come from?

- Two scenarios:
  1. **Individual-level data:** Observe D_i for each consumer
     - Survey data, loyalty card data
  2. **Market-level data:** Know distribution of D_i in each market
     - Census data, Current Population Survey
     - This is more common in practice
- pyblp handles both cases

How do we estimate this model?

- Caveat: this is one algorithm, many variations exist
- **Two-step procedure:**
  1. **Estimate γ and δ_{jt}:**
     - Maximum likelihood with δ_{jt} as product fixed effects
     - Jointly estimates demographic coefficients γ
  2. **Recover β and α:**
     - Run 2SLS on $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$
     - Need instruments because price is endogenous
- **Key insight:** First step gives us δ_{jt}, then step 2 is linear regression with endogeneity (which we know how to handle!)

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Elasticities with demographic interactions

- **Elasticities with heterogeneous consumers:**

$$\eta_{jkt} = \begin{cases} \frac{p_{jt}}{s_{jt}} \sum_i \alpha_i s_{ijt}(1 - s_{ijt}) & \text{if } j = k \\ -\frac{p_{kt}}{s_{jt}} \sum_i \alpha_i s_{ijt} s_{ikt} & \text{otherwise} \end{cases}$$

- s_{ijt}: probability that consumer i purchases product j
- **Key insight:** Cross-elasticity now depends on $\sum_i s_{ijt} s_{ikt}$
  - Products that attract *similar consumers* have higher cross-elasticity
  - BMW and Mercedes attract similar (high-income) consumers $\Rightarrow$ closer substitutes
- Correlation in μ_{ijt} and μ_{ikt} drives realistic substitution patterns

Classic example: Nevo (2001) - Cereal demand

- "Measuring Market Power in the Ready-to-Eat Cereal Industry"
- Estimated demand for cereal using demographic interactions
- Key findings:
  - High-income households less price-sensitive
  - Families with children prefer kid-targeted cereals
  - Without demographics: wrong substitution patterns
- Result: Basic logit would underestimate markups!

Limitations: Demographics vs Mixed Logit

- **Demographics model** (what we use):
  - Heterogeneity only from observed characteristics
  - IIA still holds within demographic groups
  - Simple and tractable
- **Mixed logit / random coefficients** (advanced):
  - Heterogeneity from unobserved factors too
  - Fully relaxes IIA
  - Computationally intensive (requires simulation)
- For this course: demographics are sufficient
- Know that mixed logit exists for research

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What is pyblp?

- Python package for demand estimation
- Conlon & Gortmaker (2020), "Best Practices for Differentiated Products Demand Estimation with PyBLP"
- Handles:
    - Basic logit and logit with demographics
    - Instrumental variables
    - Standard errors
    - Post-estimation (elasticities, markups, etc.)

Installing and importing packages

- **Install once** (in terminal):

```
pip install pyblp numpy pandas
```

- **Import at top of every script:**

```
import pyblp        # demand estimation
import numpy as np # numerical arrays (np.array, np.zeros, etc.)
import pandas as pd # data frames (like Excel spreadsheets)
```

- np and pd are standard abbreviations

pyblp workflow

1. **Set up data:** products, markets, shares, characteristics
2. **Define formulation:** which variables, which IVs
3. **Create problem:** combine data and formulation
4. **Solve:** estimate the model
5. **Extract results:** coefficients, standard errors, elasticities

Let's walk through each step with car data

Step 1: Load and inspect data

- Load the BLP automobile data
- Key columns: `market_ids`, `shares`, `prices`, `hpwt`, `air`, `mpd`, `space`, `demand_instruments0`, ...

```
import pyblp
import pandas as pd

product_data = pd.read_csv(pyblp.data.BLP_PRODUCTS_LOCATION)
```

What data do we need?

- **Product data** (for each product j in market t):
  - Market share s_{jt}
  - Price p_{jt}
  - Characteristics x_{jt} (size, HP, fuel efficiency, etc.)
  - Instruments z_{jt} (BLP IVs, cost shifters)
- **Demographic data** (optional, for interactions):
  - Census data on income, family size, age by market
  - Or micro-level survey data

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The BLP automobile data

market_ids	shares	prices	hpwt	air	mpd	demand_instruments0	demand_instruments1
1971	0.0012	4.94	0.52	0	1.89	2.31	0.48
1971	0.0011	5.52	0.49	0	1.94	2.28	0.51
1971	0.0006	7.11	0.47	0	1.72	2.15	0.53
1971	0.0038	4.30	0.43	0	2.45	1.98	0.57
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

- 2,217 products across 20 years (1971–1990)
- BLP instruments: sums of rivals' characteristics

Model 1: Basic logit specification

- **Utility:**

$$u_{ijt} = \beta_1 \cdot \mathrm{hpwt}_{jt} + \beta_2 \cdot \mathrm{air}_{jt} + \beta_3 \cdot \mathrm{mpd}_{jt} + \beta_4 \cdot \mathrm{space}_{jt} + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

- **Parameters to estimate:**
  - $\beta_1, \beta_2, \beta_3, \beta_4$: tastes for characteristics
  - α: price sensitivity (expect $\alpha < 0$)
- **Note:** All consumers have the same preferences (no heterogeneity)

Estimating Model 1

- Define the formulation and solve

```
# Define formulation: characteristics + price (no constant)
formulation = pyblp.Formulation('0 + hpwt + air + mpd + space + prices
    ')

# Create problem (pyblp detects instruments automatically)
problem = pyblp.Problem(formulation, product_data)

# Estimate using 2SLS
results = problem.solve()
```

Model 1: Results

Variable	**Coefficient**	**Std. Error**
HP/Weight	-1.251	(0.623)
Air conditioning	1.773	(0.194)
Miles per dollar	-1.245	(0.047)
Space	-1.643	(0.097)
Price (α)	-0.225	(0.017)

- $\hat{\alpha} = -0.225 < 0$ (higher price $\rightarrow$ lower utility)
- Some coefficients have unexpected signs (omitted variable bias without constant)

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Agent data: Consumer demographics

- pyblp uses “agent data” for demographic draws:

market_ids	**weights**	**income**
1971	0.0005	1.10
1971	0.0007	0.45
1971	0.0005	1.27
1971	0.0007	0.23
⋮	⋮	⋮

- **weights**: Integration weight for each draw (sum to 1 per market)
- **income**: Income in \$100,000s (e.g., 1.10 = \$110k)
- pyblp integrates over these draws to compute market shares

Model 2: Adding demographic interactions

- **Utility with income interactions:**

$$u_{ijt} = \delta_{jt} + \underbrace{(\pi_{\text{inc}} \cdot \text{income}_i) \cdot p_{jt}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- Where $\delta_{jt} = \beta_1 \cdot \text{hpwt}_{jt} + \ldots + \alpha_0 \cdot p_{jt} + \xi_{jt}$
- **New parameter:** π_{inc} (income $\times$ price)
    - If $\pi_{\text{inc}} > 0$: high-income less price-sensitive
- **Linear:** $\beta_1, \ldots, \alpha_0$ (in δ_{jt}) **Nonlinear:** π_{inc} (in μ_{ijt})

Estimating Model 2

```python
# Scale income to $100,000s for numerical stability
agent_data['income'] = agent_data['income'] / 100

X1 = pyblp.Formulation('0 + hpwt + air + mpd + space + prices')
X2 = pyblp.Formulation('0 + prices')  # interacts with demographics
agent_formulation = pyblp.Formulation('0 + income')

problem = pyblp.Problem(
    (X1, X2), product_data, agent_formulation, agent_data
)
results = problem.solve(sigma=0, pi=np.array([[0.5]]))
```

Understanding the solve command

- `results = problem.solve(sigma=0, pi=np.array([[0.5]]))`

- **numpy** (`np`): Python library for numerical arrays
  - `np.array([[0.5]])` creates a matrix with value 0.5

- **sigma**: Standard deviations of random coefficients
  - `sigma=0` means **no random coefficients**
  - We use demographics only, not full mixed logit
  - (Mixed logit is beyond scope of this course)

- **pi**: Demographic interaction parameters (π)
  - Starting values for the optimizer (initial guess)
  - pyblp finds the best values via GMM

- **results**: Object containing all estimation output

Model 2: Results

Parameter	Estimate	Interpretation
Linear parameters (in δ_{jt}):		
α_0 (price)	-0.728	Base price sensitivity
Demographic interaction (in μ_{ijt}):		
π_{inc} (income $\times$ price)	$+0.014$	High-income less sensitive

- Effective price coef: $\alpha_0 + \pi_{\text{inc}} \cdot \text{income}_i$
- Income $=$ \$50k: $-0.728 + 0.014 \times 0.5 = -0.721$
- Income $=$ \$200k: $-0.728 + 0.014 \times 2.0 = -0.700$ (less price-sensitive)

Elasticity matrix example

	Prod 1	**Prod 2**	**Prod 3**	**Prod 4**	**Prod 5**
Prod 1	-3.5	0.02	0.03	0.01	0.02
Prod 2	0.02	-4.1	0.03	0.02	0.03
Prod 3	0.01	0.02	-2.8	0.01	0.02
Prod 4	0.02	0.03	0.02	-3.9	0.02
Prod 5	0.01	0.02	0.01	0.01	-3.2

- **Diagonal (red):** Own-price elasticities (negative, $|\eta| > 1$)
- **Off-diagonal:** Cross-price elasticities (positive for substitutes)
- Entry (j, k): % change in s_j from 1% increase in p_k

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Post-estimation: Markups

- Recall: Lerner index $L = (p - MC)/p = 1/|\varepsilon|$

- Can recover markups from elasticities:

$$\text{markup}_j = \frac{p_j - mc_j}{p_j} = \frac{1}{|\eta_{jj}|}$$

- pyblp assumes Nash-Bertrand pricing

- Multi-product firms internalize substitution between own products

Worked example: Interpret this output

- pyblp gives you this output:

Variable	**Estimate**	**Std. Error**
Constant	3.5	0.8
HP/weight	1.2	0.4
Air cond.	0.9	0.3
Price	0.05	0.02

- **Question:** Something is wrong. What is it?

Take 1 minute to identify the problem.

Worked example: Interpret this output (solution)

- **Problem:** Price coefficient is positive (+0.05)!
  - Higher prices $\rightarrow$ higher utility. Economically nonsensical.
- **Why might this happen?**
  1. Weak instruments: IVs don't predict prices well
  2. Wrong signs: Maybe you forgot a negative sign somewhere
  3. Endogeneity still present: IVs are correlated with ξ
- **Fix:** Check first-stage F-stat, try different IVs, verify data

Common pyblp errors and how to fix them

- **"Singular matrix" error:**
  - Likely: collinear variables in formulation
  - Fix: Remove redundant variables
- **Price coefficient positive or near zero:**
  - Likely: weak instruments
  - Fix: Check first-stage F-stat; try different IVs
- **Huge standard errors:**
  - Likely: weak identification or too few observations
  - Fix: Aggregate to fewer markets; add more IVs

How to check your results make sense

1. **Price coefficient negative?** (Required!)
2. **Elasticities reasonable?**
   - Own-price: typically -2 to -10 for durables
   - Cross-price: positive but small
3. **Markups reasonable?**
   - Cars: typically 15-30%
   - Cereal: typically 30-50%
4. **First-stage F-stat > 10?** (IV relevance)
5. **Coefficients stable across specifications?**

This prepares you for HW1

- HW1 asks you to:
  1. Estimate basic logit (like Model 1)
  2. Estimate with demographics (like Model 2)
  3. Compute elasticities
  4. Interpret your results
- Key differences in HW1:
  - Uses cereal data (not cars)
  - Simpler: just sugar + price (not 5 characteristics)
  - Same structure: Model 1 $\rightarrow$ Model 2

Key Points

1. **Elasticities**: Own $= \alpha p_{jt}(1 - s_{jt})$; Cross $= -\alpha p_{kt} s_{kt} \quad (\alpha < 0)$
2. **IIA problem**: Cross-elasticities don't depend on product similarity
3. **Demographic interaction model**: $u_{ijt} = \delta_{jt} + \sum_{k,\ell} \gamma_{k\ell} D_{i\ell} x_{jt,k} + \varepsilon_{ijt}$
4. Demographics allow **heterogeneous preferences** and help with IIA
5. **pyblp workflow**: Data $\rightarrow$ Formulation $\rightarrow$ Problem $\rightarrow$ Solve $\rightarrow$ Interpret
6. **Key outputs**: Coefficients (check signs), standard errors, elasticities, markups
7. Price coefficient should be **negative**; OLS biases it toward zero
8. This is the foundation for HW1